

STA 5635 HW 9

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Problem 1

x_min	y_min	Expression 1	Expression 2	Expression 3
0.682800	0.521800	0.172200	0.172200	0.172200

Table 1: Minimized x and y values for the three expressions, as well as the values of the expressions.

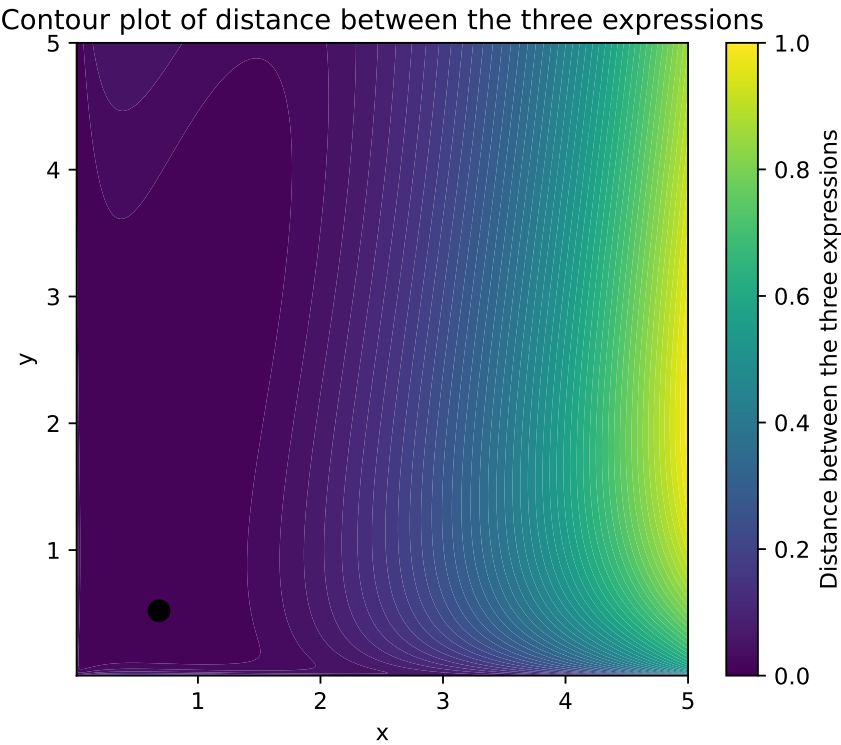


Figure 1: Distance between the three equations and what x, y value minimizes said distance.

Problem 2

Part a

Part b

The ratio of alpha values at time 118: 0.4970

The ratio of beta values at time 118: 1.3175

Problem 3

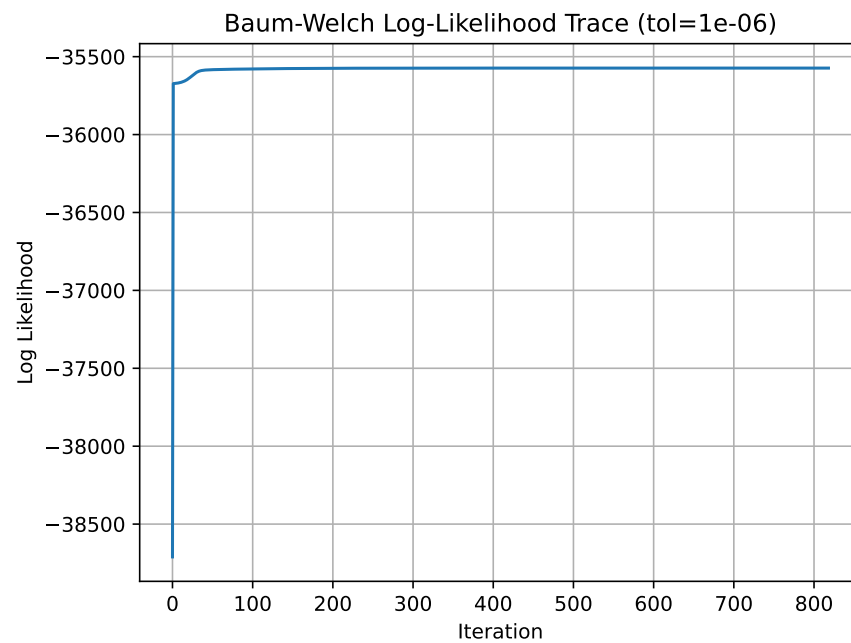


Figure 2: Log Likelihood of Baum Algorithm, showing convergence.